

ASSIGNMENT 2

FAROOQ ABDULLA

NIELIT

Online Internship on Analog and Digital VLSI Design Flow

Batch 1

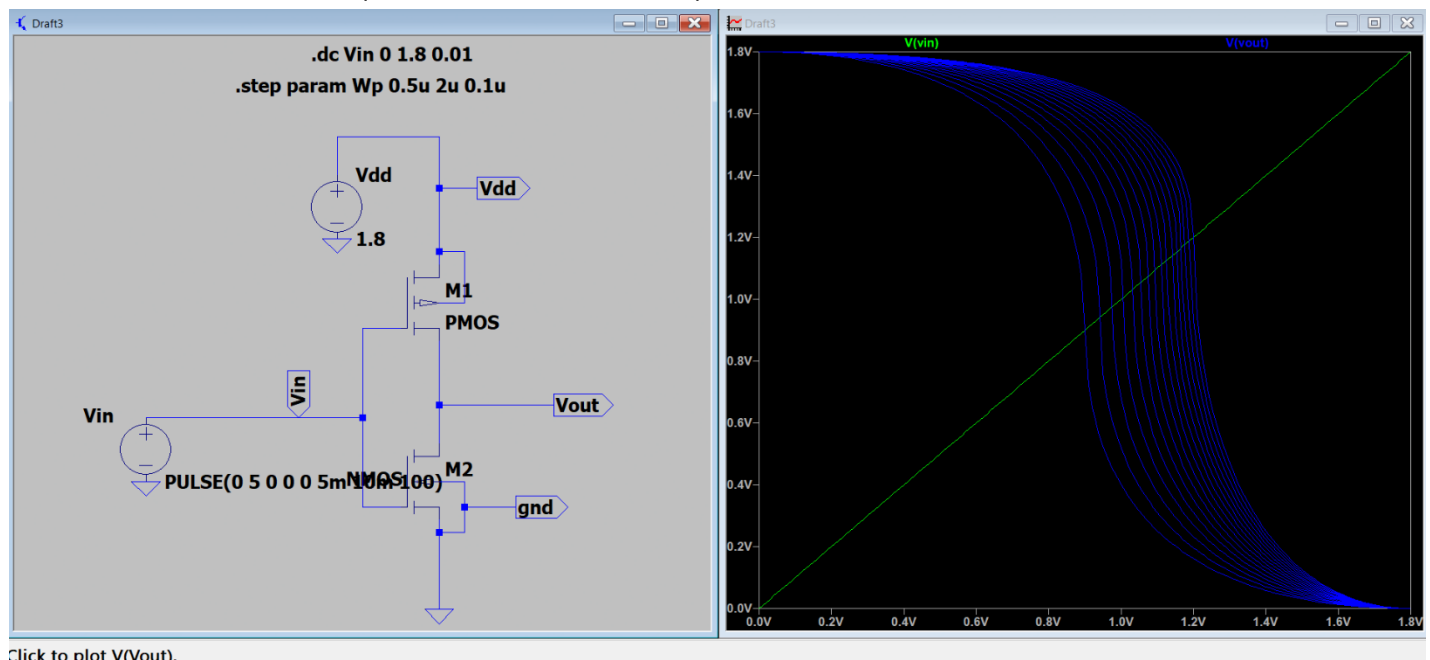
Design CMOS inverter with mid point voltage (V_m) = $V_{DD}/2$. Confirm the value using the VTC curve of the inverter. (Hint : fix length and vary width of PMOS and NMOS)

Then Find :

1. Noise Margin
2. Propagation Delay (T_{plh} and T_{phl})
3. Rise Time and Fall Time
4. Dynamic and Static Power Consumed

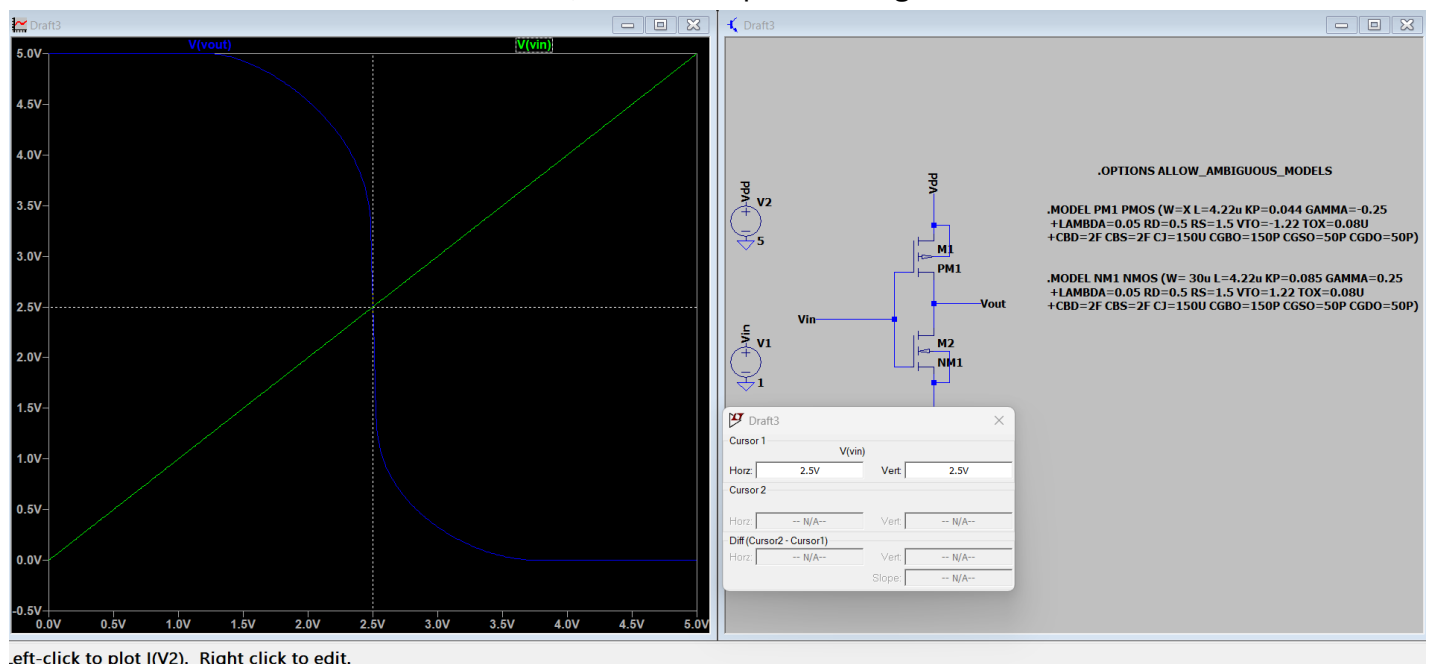
Answers and Screenshot

1. Schematic screenshot (with W/L values labeled)



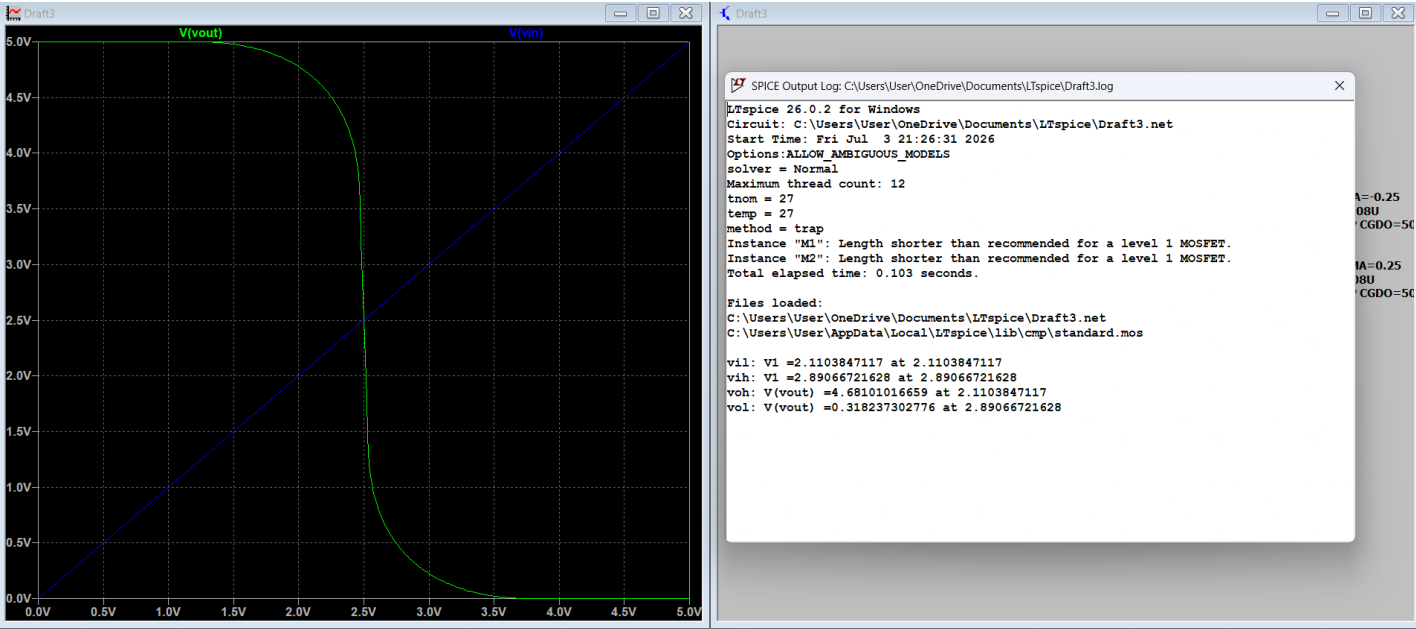
Click to plot V(Vout).

2. VTC curve screenshot with V_m marked, and the output showing $V_m \approx V_{DD}/2$

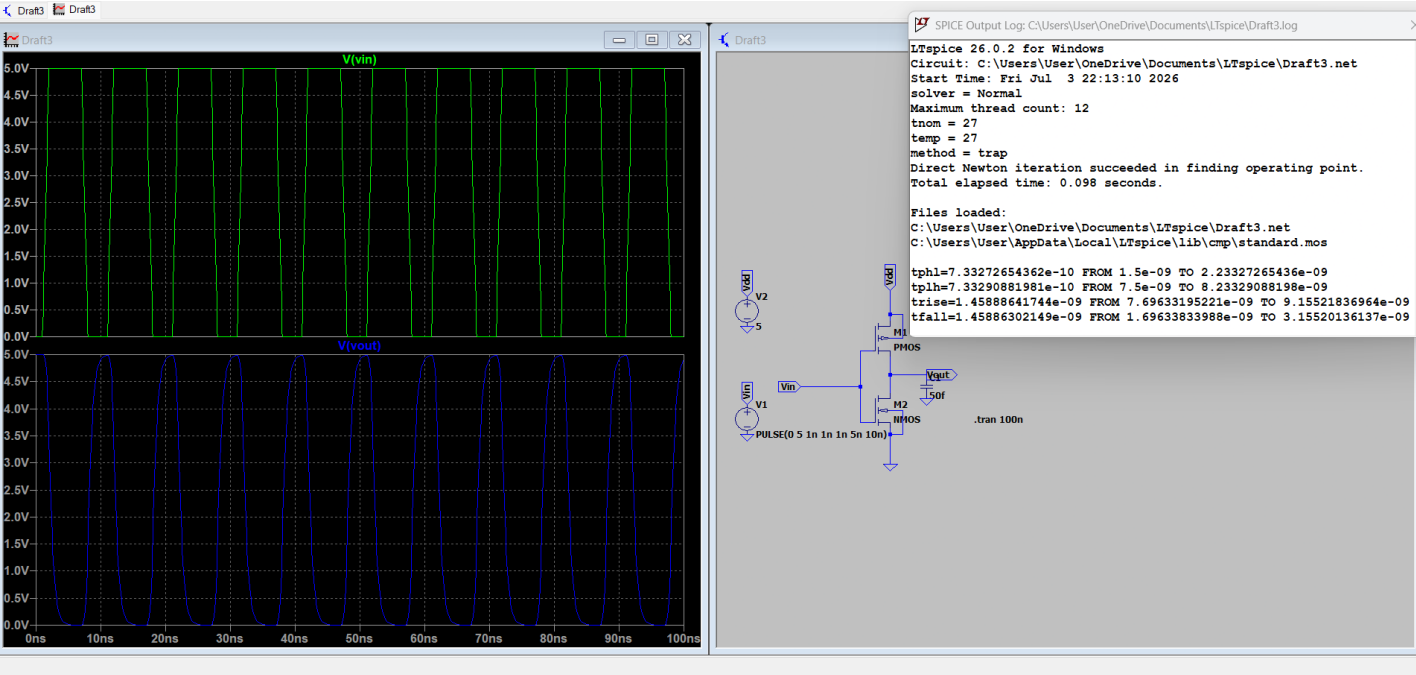


Left-click to plot I(V2). Right click to edit.

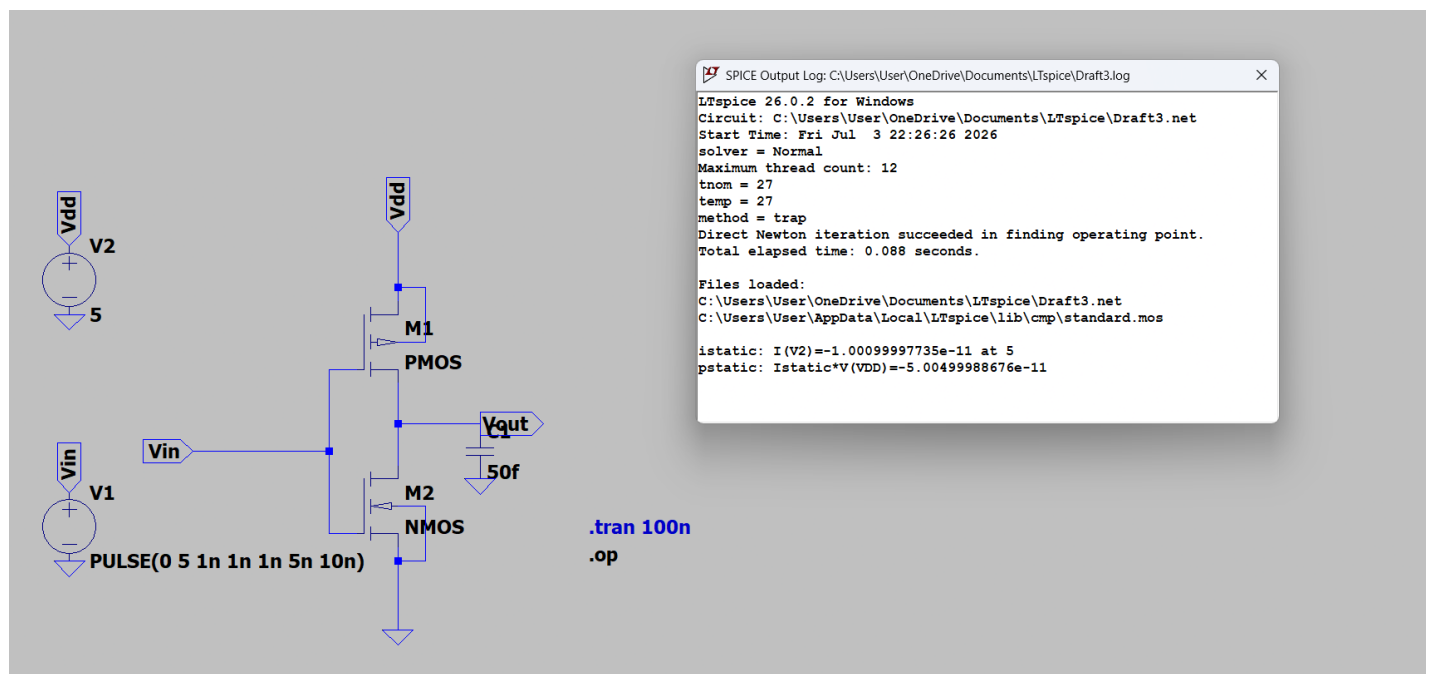
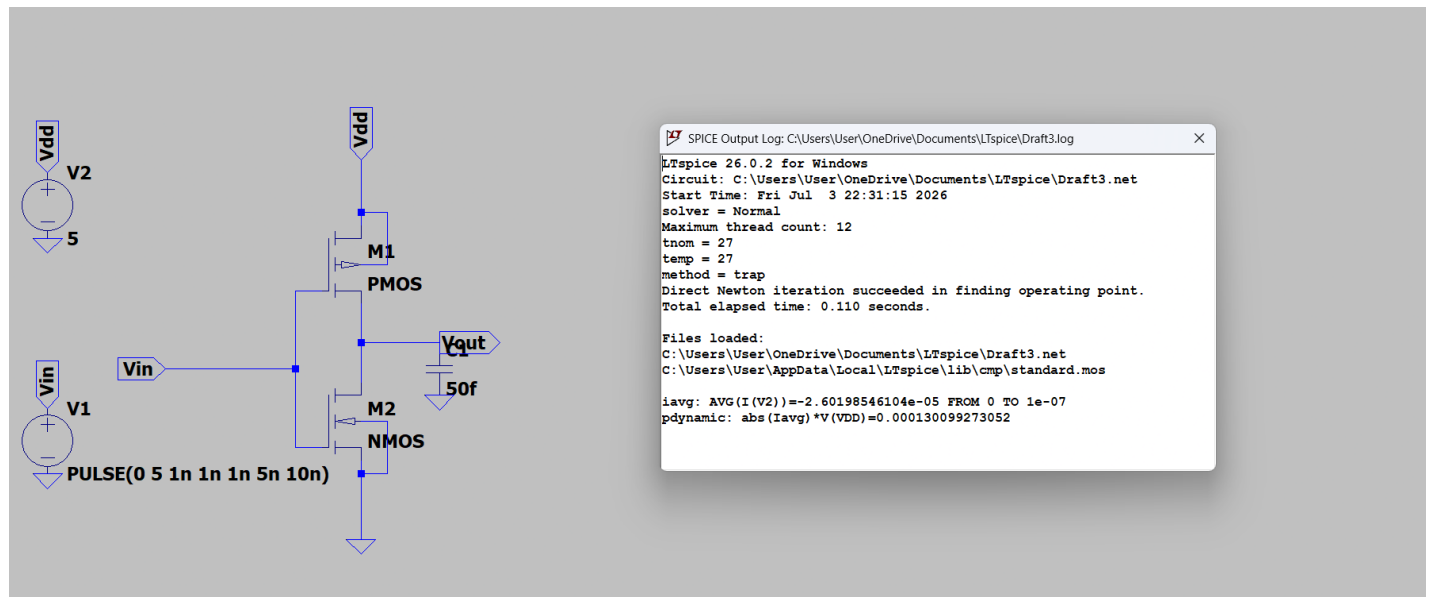
3. VTC curve annotated with VIL, VIH, VOL, VOH and computed NMH/NML



4. Transient waveform (Vin & Vout overlaid) with Tplh/Tphl to find Trise and Tfall



5. Log output for static and dynamic power



Vdd/2 =Vm	2.5v
Noise Margin High	1.921
Noise Margin Low	1.792
Tplh	7.332x10 ⁻¹⁰
TPhl	7.332x10 ⁻¹⁰
Trise	1.458x10 ⁻⁹
Tfall	1.458x10 ⁻⁹
Dynamic Power	0.000130 watts
Static Power	-5.004x10 ⁻¹¹ watts